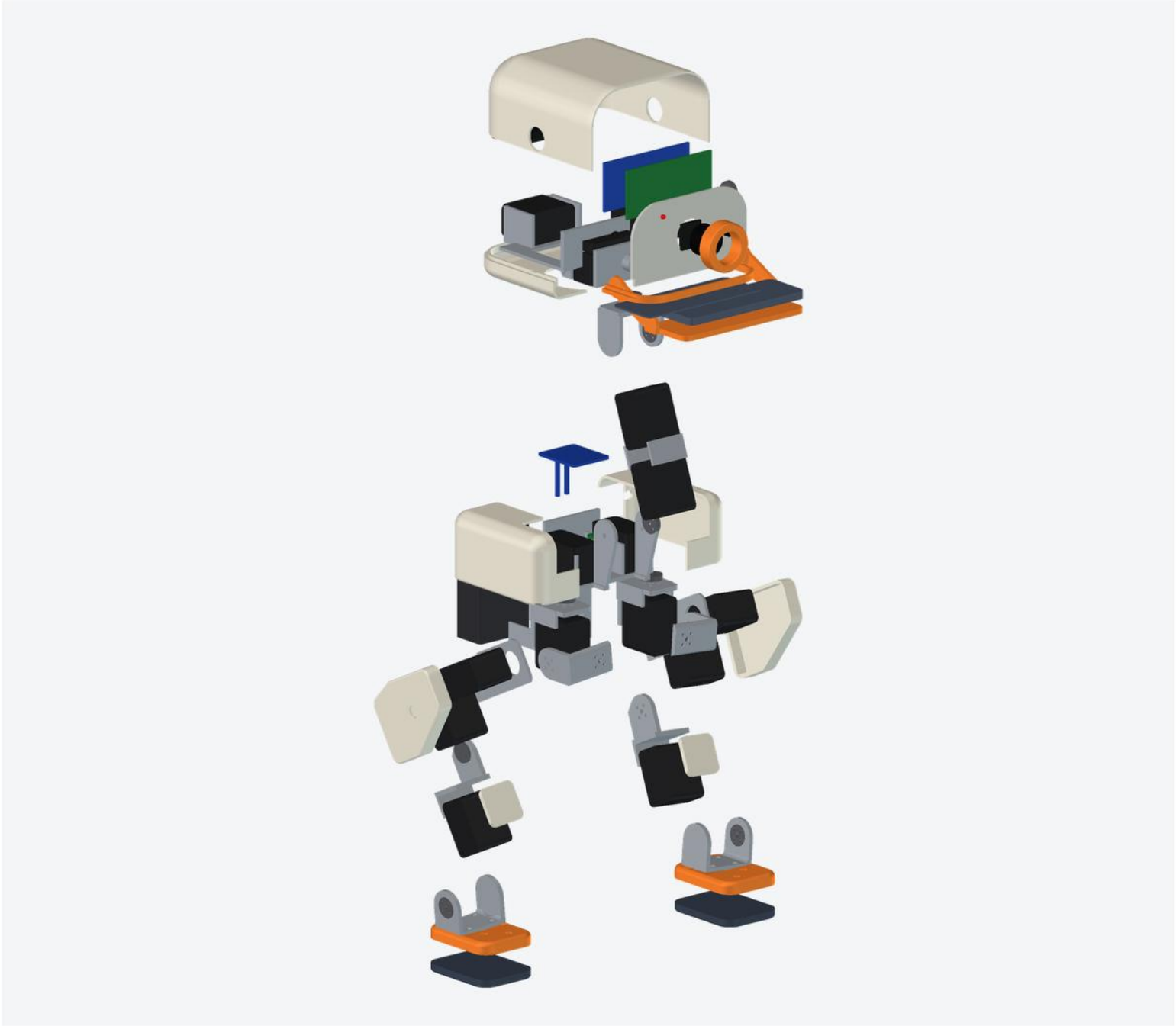
OpenDuck (Microduck benzeri) | 273 mm | ~676 g | 15 x DYNAMIXEL XL330-M288-T | 2026-10-23



Patlatilmis montaj

tum robot, HOME pozunda; eklem ofsetleri buyutuldu

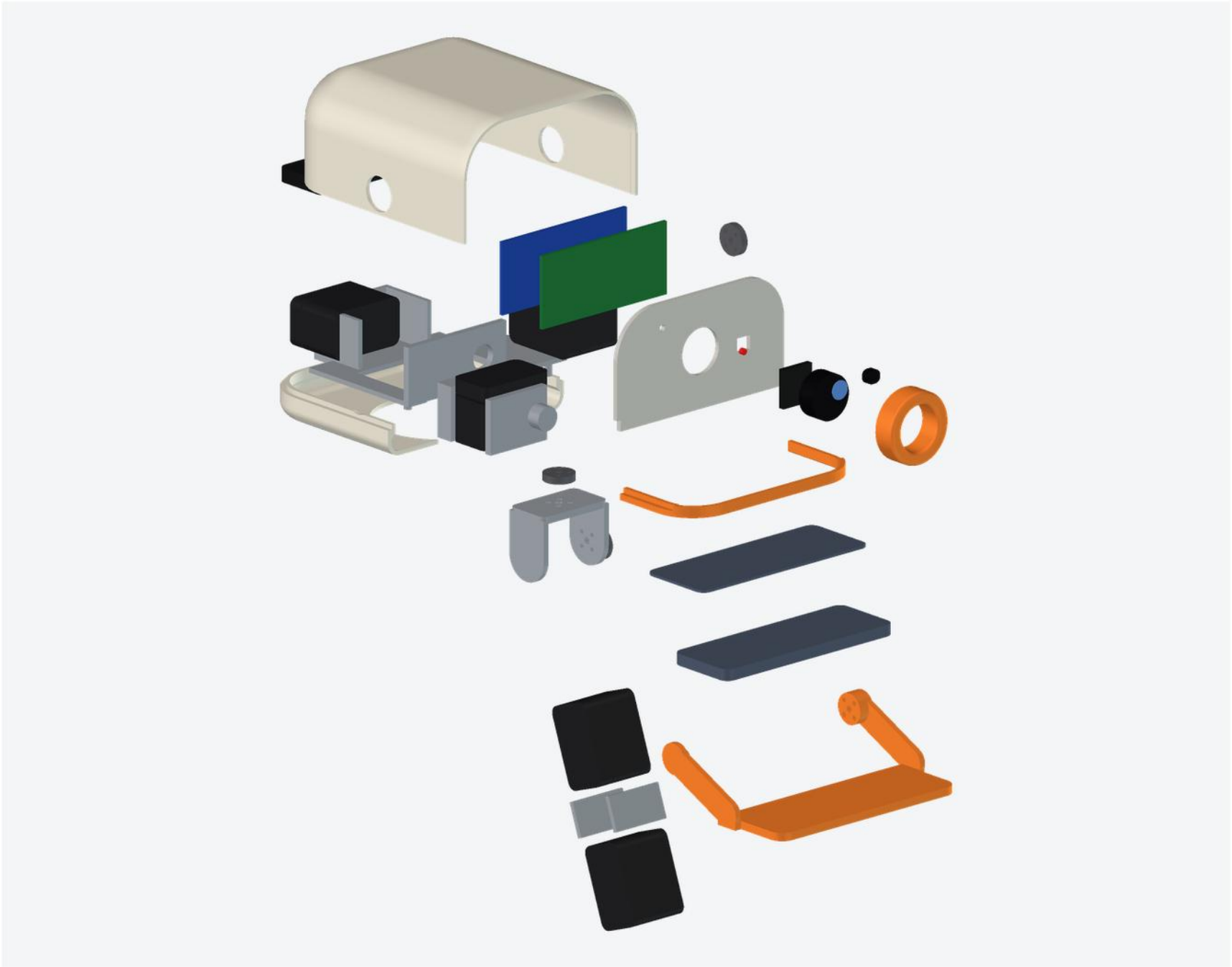


Patlatilmis montaj - yandan



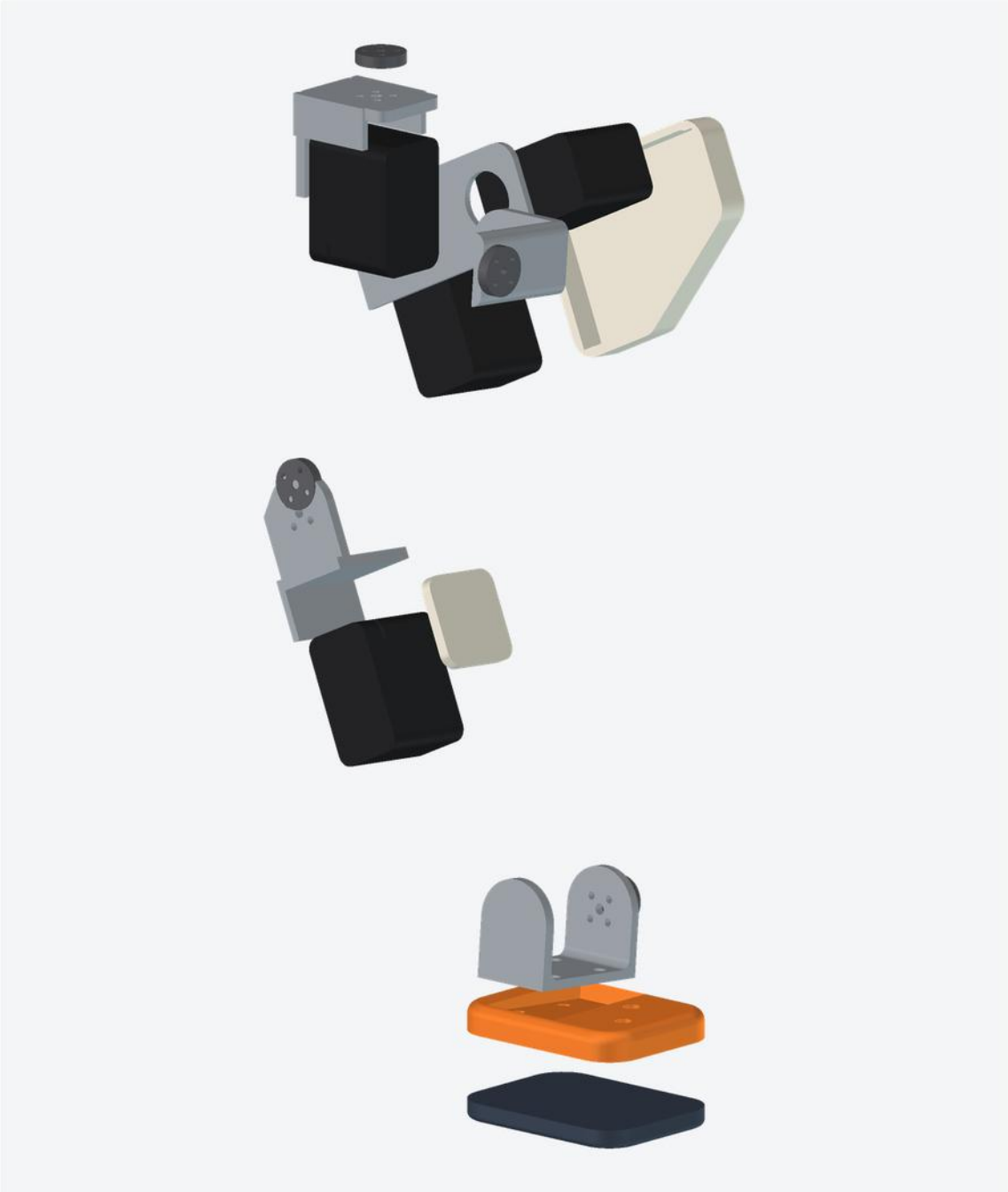
Kafa alt montaji

boyun, kafa egim/donus braketleri, kafa ici; yuz yigini goz ekseni boyunca, gaga parcalari (ust gaga, ust ped, alt ped, alt gaga) asagi dogru



Sol bacak alt montaji

kalca yaw -> roll -> pitch -> diz -> ayak bilegi; sag bacak ayna



Malzeme listesi

Satin alinacak parcalar

- 15 x ROBOTIS DYNAMIXEL XL330-M288-T - bacaklar 2x5, boyun+kafa 4, gaga 1 (ID 10-14, 20-24, 30-34)
- 1 x Radxa Zero 3W (RK3566) veya Raspberry Pi Zero 2 W - kafada, dik
- 1 x Robot HAT (Pollen elec_RPI_Robot_HAT, Apache-2.0) veya esdegeri - Dynamixel TTL, ses, 5 V
- 1 x IMX219 kamera + M12 lens - goz
- 1 x VL53L5CX 8x8 ToF - 'lidar'
- 1 x LSM6DSV16X IMU karti - govde
- 1 x NP-F550 2S Li-ion (7.4 V, 2600 mAh) + yuva konnektoru - servolara dogrudan
- 1 x Hoparlor 25x35 mm, MEMS mikrofon
- 1 x M2 vida seti, isil ek somun, kablolar - bacak horn baglantilari 2x20 + boyun catali 4 vida; ayak bilegi U braketi 2x4 havsa basli vida + 8 isil ek; batarya yuvasi 3 vida + 3 isil ek

3B baski

27 baski parcasi cesidi, toplam 36 parca (sol + sag dahil); liste: katalog sayfalari ve print/PRINT.md.

Servo kimlikleri (ID)

sol bacak 20-24, sag bacak 10-14 (hip_yaw, hip_roll, hip_pitch, knee, ankle)

neck_pitch 30, head_pitch 31, head_yaw 32, head_roll 33, mouth (gaga) 34

Montaj sirasi (oneri)

0) Servolar, montajdan ONCE: her XL330'u TEK BASINA bagla (fabrikadan hepsi ID 1 gelir), Dynamixel Wizard ile ID'sini yukaridaki tabloya gore ata ve etiketle, servoyu HOME acisina sur (FCStd 'Poz' tablosu, B sutunu). Horn'u ve braketi ANCAK bundan sonra, servo HOME'dayken tak.

1) Govde: sasiye kalca yaw servolarini ve boyun servosunu (ID 30) tak; IMU kartini ve guc kartini sasiye vidala; batarya yuvasini (battery_tray) sasiye tak.

2) Her bacakta: hip_bracket + roll servosu -> hip_cradle_bracket -> uyluk (pitch + diz servosu, ic plaka, kanat) -> incik (bilek servosu, braket, ped) -> ayak (U braketi 4x M2 ile ayak govdesine, taban).

3) Boyun: iki servoyu ek plakalariyla (neck_splice) birlestir, kafa egim braketini (head_tilt_bracket) tak; yaw servosunu kafa donus braketinin (head_turn_bracket) iki parcasi arasina al: roll horn'u arka parcaya, on parcadaki 10 mm pim roll ekseninin karsi yatagina.

4) Kafa: ic cerceve + roll servosu + gaga servosu, kart/HAT, kamera, ToF, hoparlor, alt kabuk, yuz plakasi, goz halkasi, ust gaga + ust gaga pedi (TPU, ust gaganin altina); alt gaga (cene) servoya, alt gaga pedi (TPU) alt gaganin ustune; en son ust kaput.

5) Govdeyi kapat: kablolari topla, bataryayi yuvasina yerlestir, govde kabugunun sol ve sag yarilarini (torso_shell_left / torso_shell_right) sasiye tak.

baski yonunde, izometrik; renkler oneridir; adet sol + sag dahil; her kare bir STL cesidi



chassis x1

PETG (veya PLA+), 17 g

60.5 x 60.0 x 42.9 mm



battery_tray x1

PETG (veya PLA+), 10 g

32.3 x 55.0 x 41.0 mm



torso_shell_left x1

PLA, 11 g

80.9 x 41.7 x 32.2 mm



torso_shell_right x1

PLA, 11 g

80.9 x 41.7 x 32.2 mm



hip_bracket x2

PETG (veya PLA+), 5 g

27.0 x 24.0 x 30.0 mm



hip_cradle_bracket x2

PETG (veya PLA+), 3 g

29.0 x 32.0 x 19.0 mm



thigh_inner_plate x2

PETG (veya PLA+), 3 g

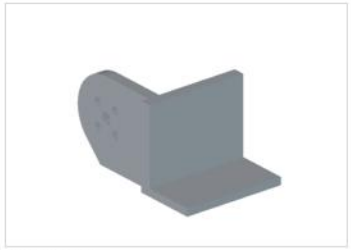
61.3 x 27.5 x 2.0 mm



thigh_wing x2

PLA, 10 g

61.3 x 47.5 x 9.5 mm



shin_bracket x2

PETG (veya PLA+), 4 g

40.5 x 26.0 x 22.0 mm



shin_pad x2

PLA, 2 g

24.0 x 24.0 x 3.0 mm



ankle_bracket x2

PETG (veya PLA+), 5 g

22.0 x 35.0 x 27.0 mm



foot_body x2

PLA, 14 g

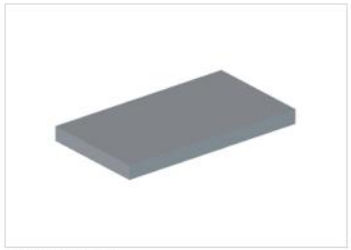
54.0 x 40.5 x 7.5 mm



sole x2

TPU 95A, 13 g

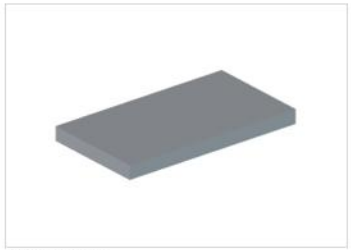
54.0 x 40.5 x 5.0 mm



neck_splice_1 x1

PETG (veya PLA+), 1 g

14.5 x 26.0 x 2.0 mm



neck_splice_2 x1

PETG (veya PLA+), 1 g

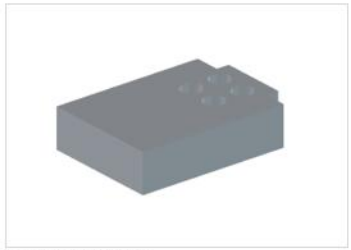
14.5 x 26.0 x 2.0 mm



head_tilt_bracket x1

PETG (veya PLA+), 5 g

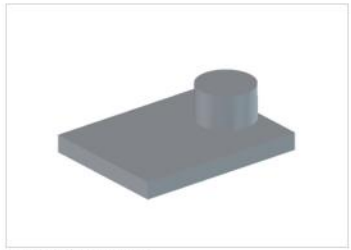
28.7 x 34.0 x 20.0 mm



head_turn_bracket_1 x1

PETG (veya PLA+), 6 g

22.5 x 32.0 x 7.9 mm



head_turn_bracket_2 x1

PETG (veya PLA+), 3 g

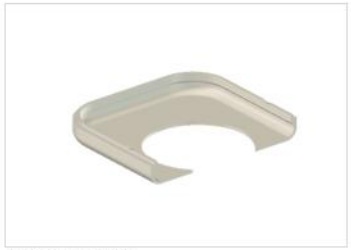
22.5 x 29.5 x 9.0 mm



head_hood x1

PLA, 38 g, destekli

105.6 x 91.8 x 47.4 mm



head_bottom_shell x1

PLA, 15 g

76.4 x 91.8 x 13.3 mm



upper_bill x1

PLA, 2 g

40.3 x 91.8 x 4.5 mm



upper_bill_pad x1

TPU 85-95A, 4 g

33.3 x 87.2 x 1.1 mm



face_plate x1

PLA, 8 g

47.8 x 87.6 x 2.0 mm



eye_ring x1

PLA, 4 g

30.0 x 30.0 x 9.5 mm

baski yonunde, izometrik; renkler oneridir; adet sol + sag dahil; her kare bir STL cesidi



head_frame x1

PETG (veya PLA+), 14 g, destekli

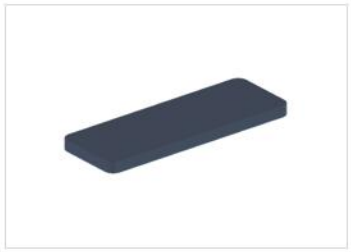
27.5 x 78.5 x 73.5 mm



lower_bill x1

PLA, 15 g, destekli

65.7 x 98.8 x 26.9 mm

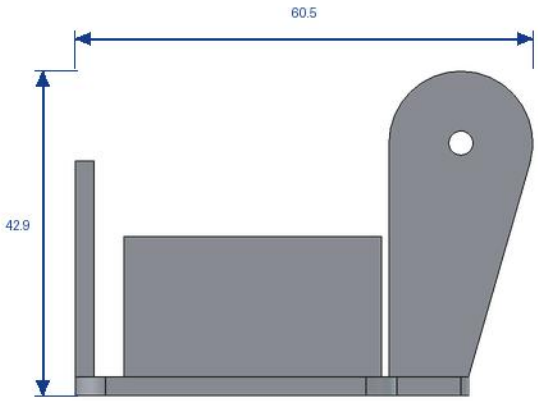


lower_bill_pad x1

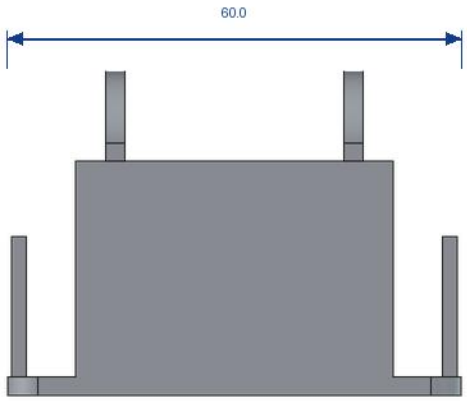
TPU 85-95A, 16 g

31.3 x 87.8 x 4.8 mm

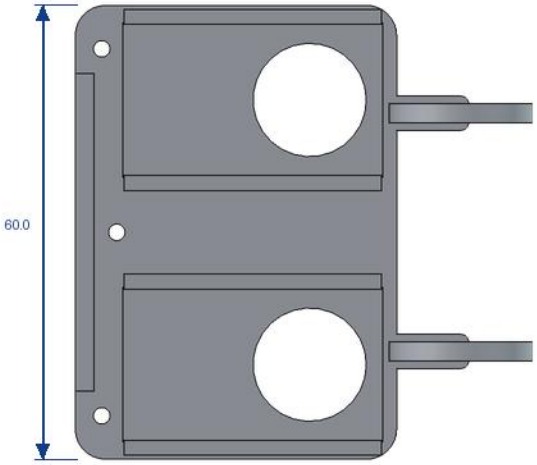
Olculer mm. Genel zarf olculeri verilmistir; delikler ve detaylar icin STEP/STL kullanin.



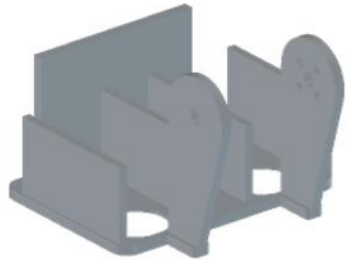
ON GORUNUS



SOLDAN GORUNUS



USTTEN GORUNUS

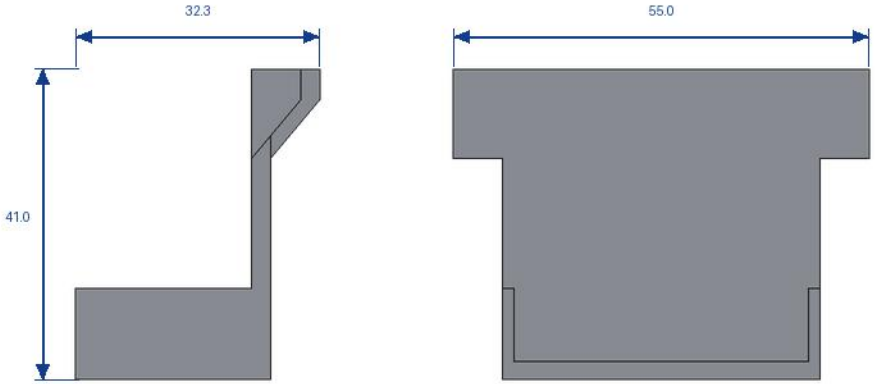


OpenDuck

Microduck benzeri ordek robot

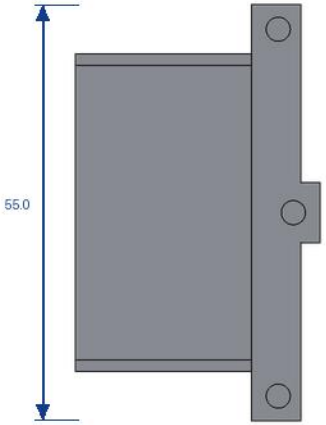
PARCA	chassis
ACIKLAMA	sasi (taban + servo kafesi + boyun catali)
ADET	1
MALZEME	PETG (veya PLA+), renk: gri
BASKI	0.2 mm katman, 4 duvar, %40 dolgu
KUTLE	16.5 g (adet basi, tahmini)
ZARF	60.5 x 60.0 x 42.9 mm
YON	otomatik (genis temas, az destek)
DESTEK	gerekmez
OLCEK	1:1 (A4, 150 dpi)
IZDUSUM	ISO E - 1. aci, baski yonunde
DOSYA	chassis.stl

Olculer mm. Genel zarf olculeri verilmistir; delikler ve detaylar icin STEP/STL kullanin.

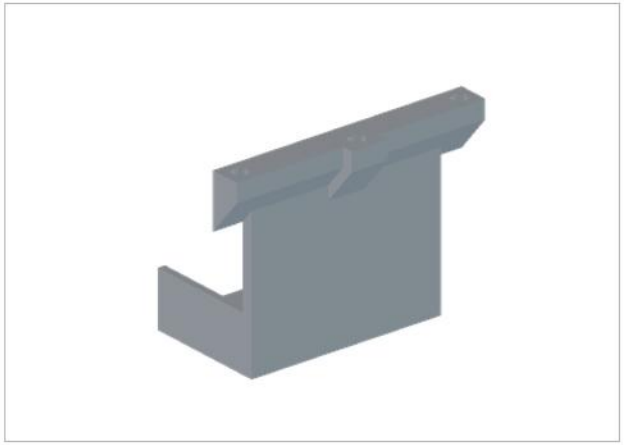


ON GORUNUS

SOLDAN GORUNUS



USTTEN GORUNUS

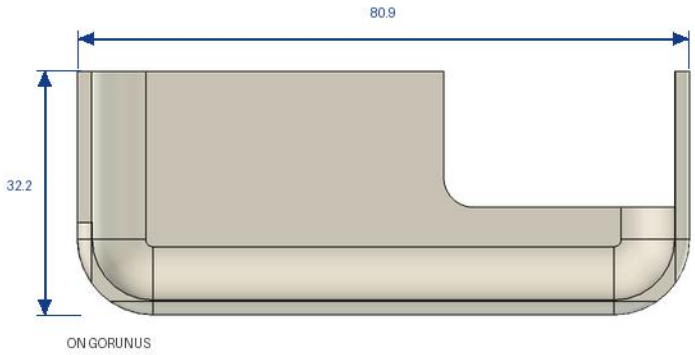


OpenDuck

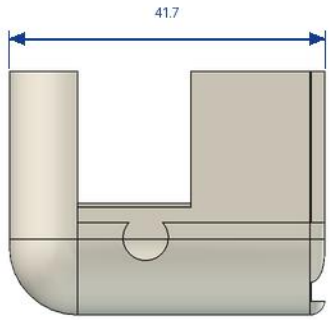
Microduck benzeri ordek robot

PARCA	battery_tray
ACIKLAMA	batarya yuvasi
ADET	1
MALZEME	PETG (veya PLA+), renk: gri
BASKI	0.2 mm katman, 4 duvar, %40 dolgu
KUTLE	10.1 g (adet basi, tahmini)
ZARF	32.3 x 55.0 x 41.0 mm
YON	otomatik (genis temas, az destek)
DESTEK	gerekmez
OLCEK	1:1 (A4, 150 dpi)
IZDUSUM	ISO E - 1. aci, baski yonunde
DOSYA	battery_tray.stl

Olculer mm. Genel zarf olculeri verilmistir; delikler ve detaylar icin STEP/STL kullanin.



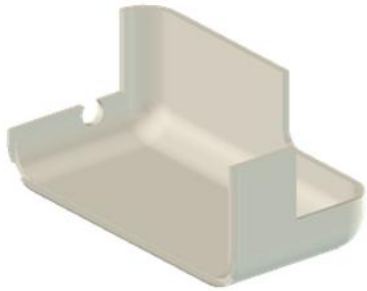
ON GORUNUS



SOLDAN GORUNUS



USTTEN GORUNUS

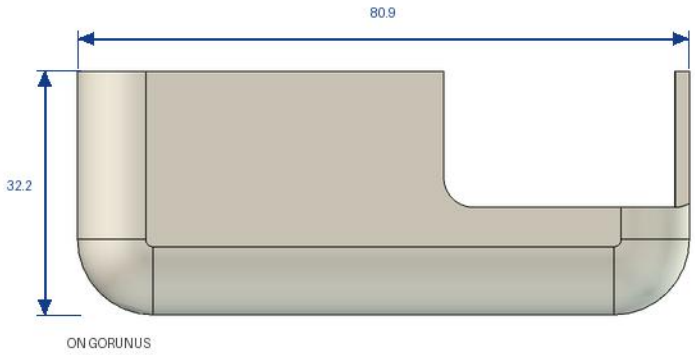


OpenDuck

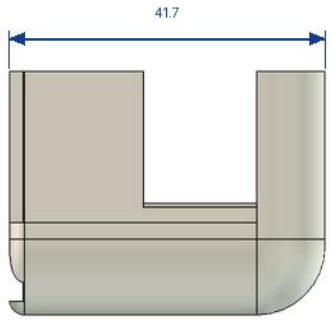
Microduck benzeri ordek robot

PARCA	torso_shell_left
ACIKLAMA	govde kabugu sol yarim
ADET	1
MALZEME	PLA, renk: krem
BASKI	0.2 mm katman, 3 duvar, %15 dolgu
KUTLE	11.4 g (adet basi, tahmini)
ZARF	80.9 x 41.7 x 32.2 mm
YON	disyan yuz tablada: desteksiz, alcak, gorunen yuz tabla yuzeyi
DESTEK	gerekmez
OLCEK	1:1 (A4, 150 dpi)
IZDUSUM	ISO E - 1. aci, baski yonunde
DOSYA	torso_shell_left.stl

Olculer mm. Genel zarf olculeri verilmistir; delikler ve detaylar icin STEP/STL kullanin.



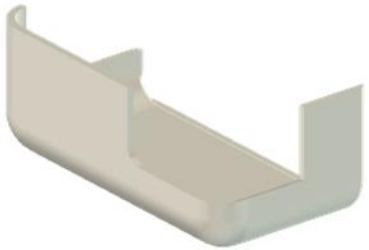
ON GORUNUS



SOLDAN GORUNUS



USTTEN GORUNUS

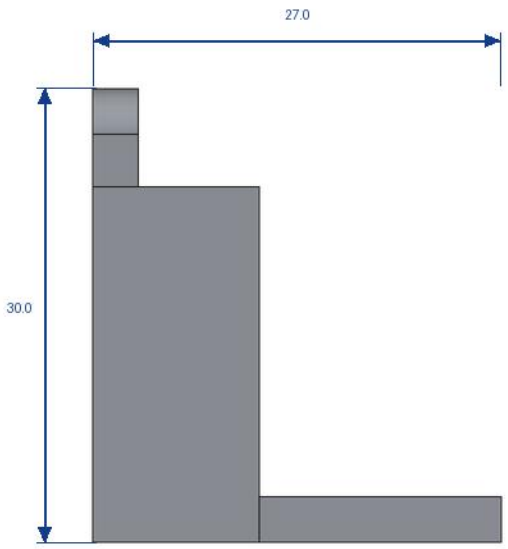


OpenDuck

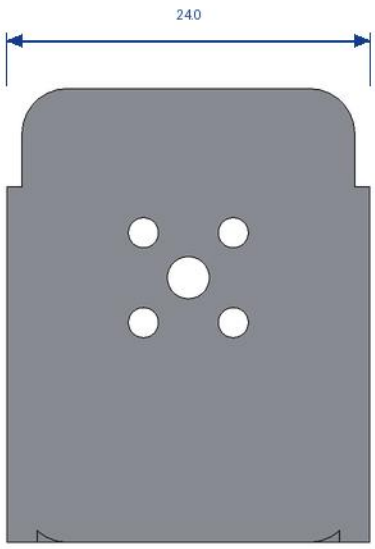
Microduck benzeri ordek robot

PARCA	torso_shell_right
ACIKLAMA	govde kabugu sag yarim
ADET	1
MALZEME	PLA, renk: krem
BASKI	0.2 mm katman, 3 duvar, %15 dolgu
KUTLE	11.4 g (adet basi, tahmini)
ZARF	80.9 x 41.7 x 32.2 mm
YON	dis yan yuz tablada: desteksiz, alcak, gorunen yuz tabla yuzeyi
DESTEK	gerekmez
OLCEK	1:1 (A4, 150 dpi)
IZDUSUM	ISO E - 1. aci, baski yonunde
DOSYA	torso_shell_right.stl

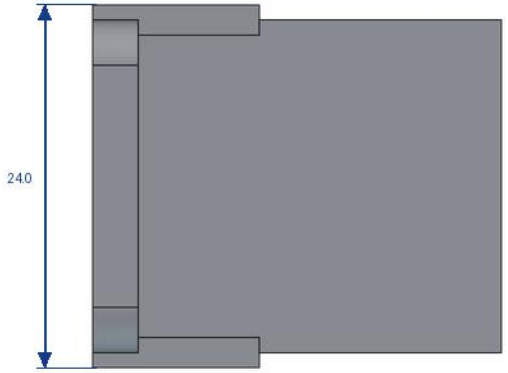
Olculer mm. Genel zarf olculeri verilmistir; delikler ve detaylar icin STEP/STL kullanin.



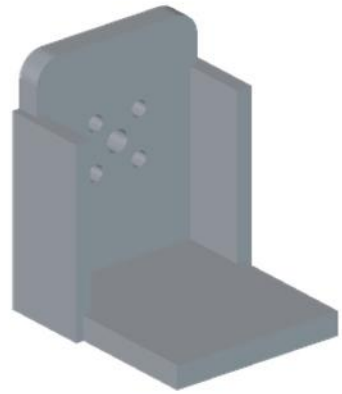
ON GORUNUS



SOLDAN GORUNUS



USTTEN GORUNUS

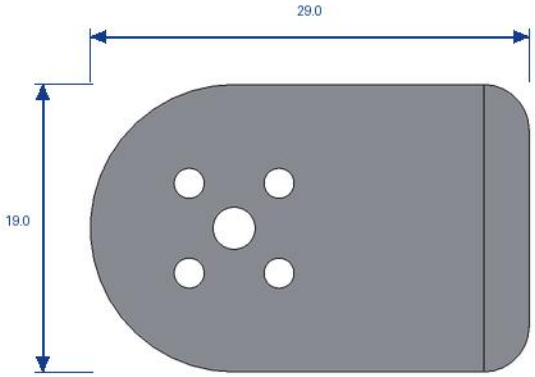


OpenDuck

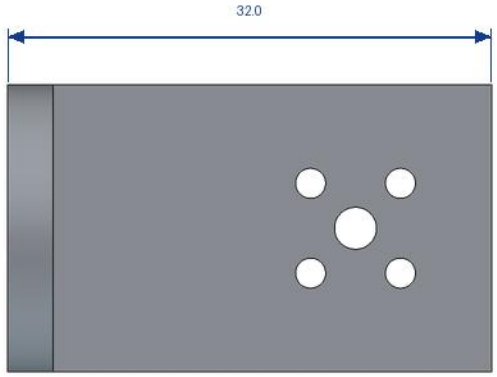
Microduck benzeri ordek robot

PARCA	hip_bracket
ACIKLAMA	kalca yaw braketi
ADET	2 (sol + sag, ayna)
MALZEME	PETG (veya PLA+), renk: gri
BASKI	0.2 mm katman, 4 duvar, %40 dolgu
KUTLE	4.8 g (adet basi, tahmini)
ZARF	27.0 x 24.0 x 30.0 mm
YON	servo arka plakasi katman duzleminde, desteksiz
DESTEK	gerekmez
OLCEK	2:1 (A4, 150 dpi)
IZDUSUM	ISO E - 1. aci, baski yonunde
DOSYA	left_hip_bracket.stl, right_hip_bracket.stl

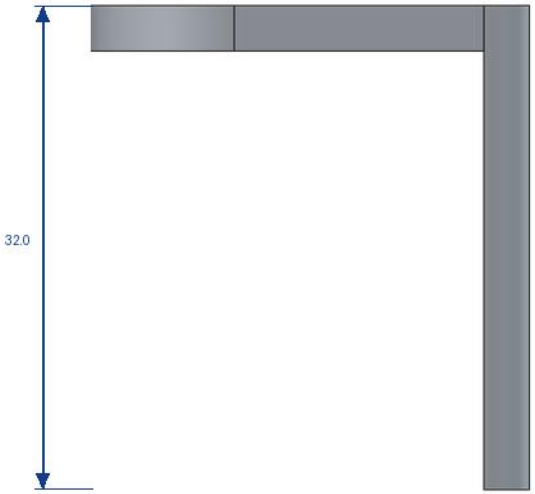
Olculer mm. Genel zarf olculeri verilmistir; delikler ve detaylar icin STEP/STL kullanin.



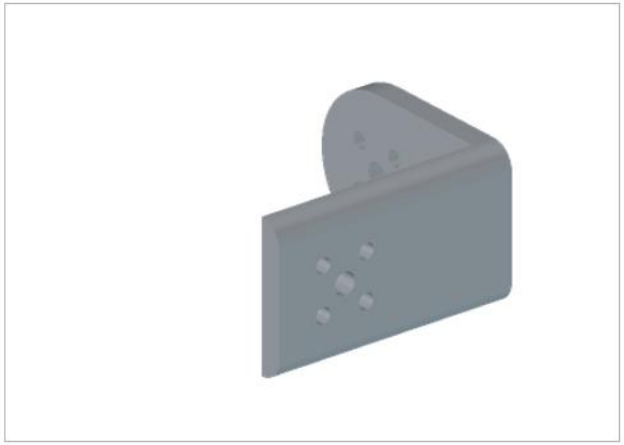
ON GORUNUS



SOLDAN GORUNUS



USTTEN GORUNUS

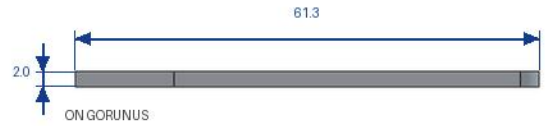


OpenDuck

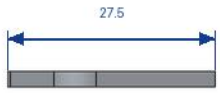
Microduck benzeri ordek robot

PARCA	hip_cradle_bracket
ACIKLAMA	kalca roll-pitch braketi
ADET	2 (sol + sag, ayna)
MALZEME	PETG (veya PLA+), renk: gri
BASKI	0.2 mm katman, 4 duvar, %40 dolgu
KUTLE	3.2 g (adet basi, tahmini)
ZARF	29.0 x 32.0 x 19.0 mm
YON	L profili katman duzleminde, kollar bukumden ayrilmaz; 5 mm kenar (brim)
DESTEK	gerekmez
OLCEK	2:1 (A4, 150 dpi)
IZDUSUM	ISO E - 1. aci, baski yonunde
DOSYA	left_hip_cradle_bracket.stl, right_hip_cradle_bracket.stl

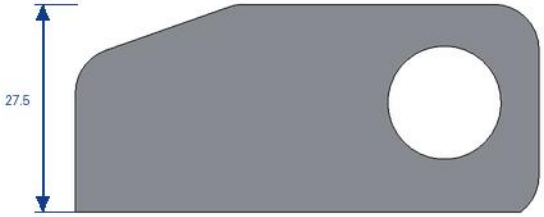
Olculer mm. Genel zarf olculeri verilmistir; delikler ve detaylar icin STEP/STL kullanin.



ON GORUNUS



SOLDAN GORUNUS



USTTEN GORUNUS

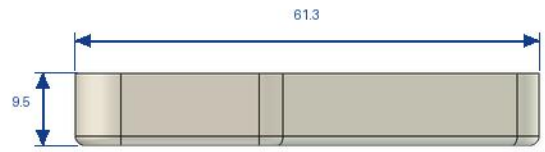


OpenDuck

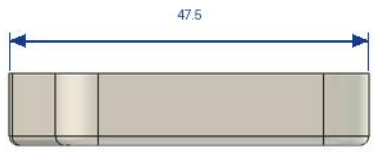
Microduck benzeri ordek robot

PARCA	thigh_inner_plate
ACIKLAMA	uyluk ic plakasi
ADET	2 (sol + sag, ayna)
MALZEME	PETG (veya PLA+), renk: gri
BASKI	0.2 mm katman, 4 duvar, %40 dolgu
KUTLE	3.1 g (adet basi, tahmini)
ZARF	61.3 x 27.5 x 2.0 mm
YON	otomatik (genis temas, az destek)
DESTEK	gerekmez
OLCEK	1:1 (A4, 150 dpi)
IZDUSUM	ISO E - 1. aci, baski yonunde
DOSYA	left_thigh_inner_plate.stl, right_thigh_inner_plate.stl

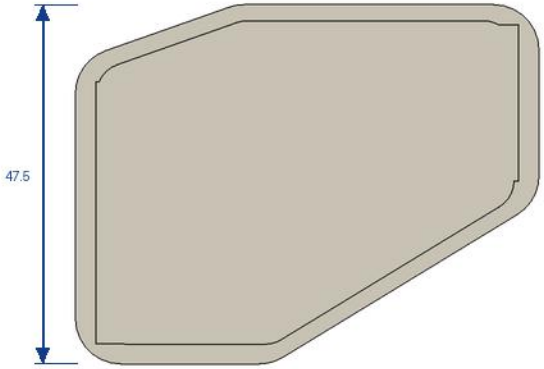
Olculer mm. Genel zarf olculeri verilmistir; delikler ve detaylar icin STEP/STL kullanin.



ON GORUNUS



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USTTEN GORUNUS

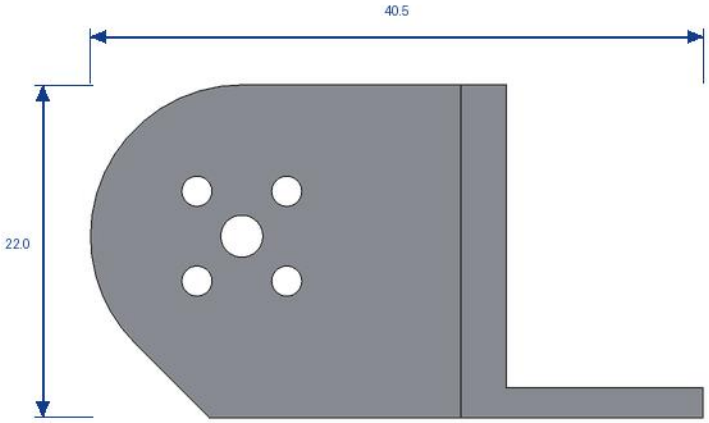


OpenDuck

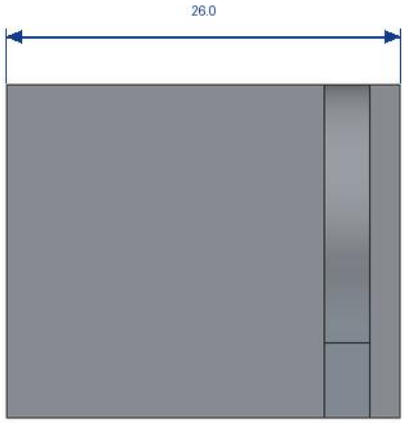
Microduck benzeri ordek robot

PARCA	thigh_wing
ACIKLAMA	uyluk kanat kabugu
ADET	2 (sol + sag, ayna)
MALZEME	PLA, renk: krem
BASKI	0.2 mm katman, 3 duvar, %15 dolgu
KUTLE	9.9 g (adet basi, tahmini)
ZARF	61.3 x 47.5 x 9.5 mm
YON	otomatik (genis temas, az destek)
DESTEK	gerekmez
OLCEK	1:1 (A4, 150 dpi)
IZDUSUM	ISO E - 1. aci, baski yonunde
DOSYA	left_thigh_wing.stl, right_thigh_wing.stl

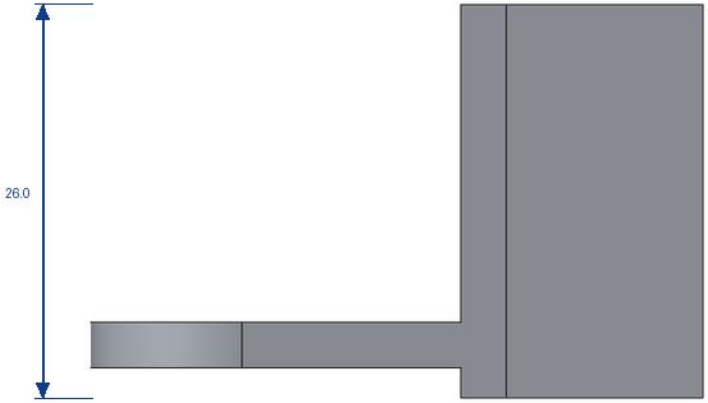
Olculer mm. Genel zarf olculeri verilmistir; delikler ve detaylar icin STEP/STL kullanin.



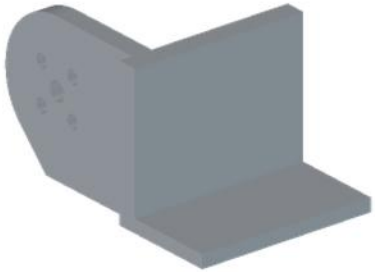
ON GORUNUS



SOLDAN GORUNUS



USTTEN GORUNUS

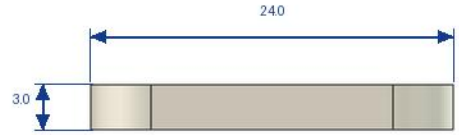


OpenDuck

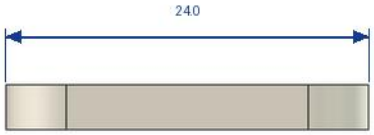
Microduck benzeri ordek robot

PARCA	shin_bracket
ACIKLAMA	incik braketi
ADET	2 (sol + sag, ayna)
MALZEME	PETG (veya PLA+), renk: gri
BASKI	0.2 mm katman, 4 duvar, %40 dolgu
KUTLE	4.2 g (adet basi, tahmini)
ZARF	40.5 x 26.0 x 22.0 mm
YON	otomatik (genis temas, az destek)
DESTEK	gerekmez
OLCEK	2:1 (A4, 150 dpi)
IZDUSUM	ISO E - 1. aci, baski yonunde
DOSYA	left_shin_bracket.stl, right_shin_bracket.stl

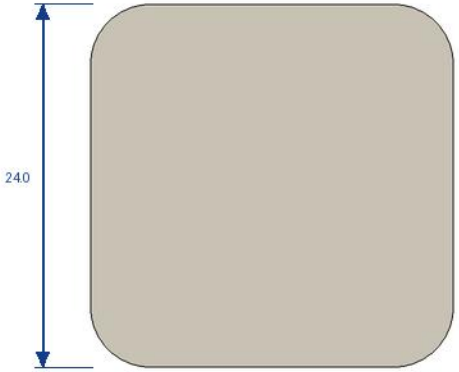
Olculer mm. Genel zarf olculeri verilmistir; delikler ve detaylar icin STEP/STL kullanin.



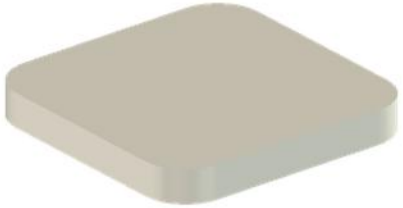
ON GORUNUS



SOLDAN GORUNUS



USTTEN GORUNUS

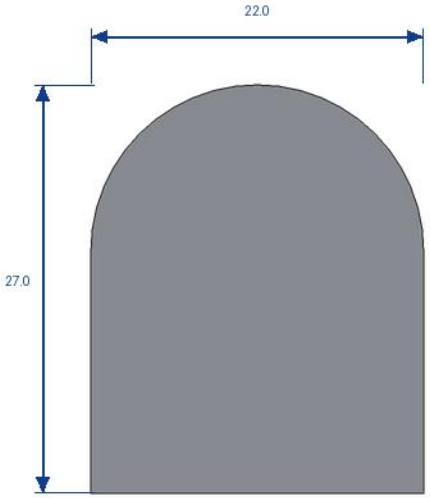


OpenDuck

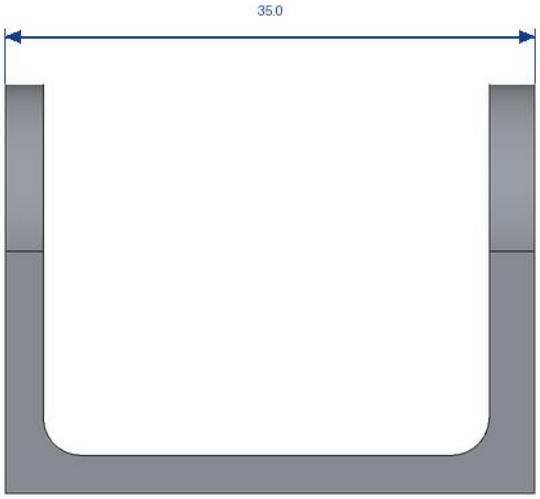
Microduck benzeri ordek robot

PARCA	shin_pad
ACIKLAMA	diz/incik pedi
ADET	2 (sol + sag, ayna)
MALZEME	PLA, renk: krem
BASKI	0.2 mm katman, 3 duvar, %15 dolgu
KUTLE	1.9 g (adet basi, tahmini)
ZARF	24.0 x 24.0 x 3.0 mm
YON	otomatik (genis temas, az destek)
DESTEK	gerekmez
OLCEK	2:1 (A4, 150 dpi)
IZDUSUM	ISO E - 1. aci, baski yonunde
DOSYA	left_shin_pad.stl, right_shin_pad.stl

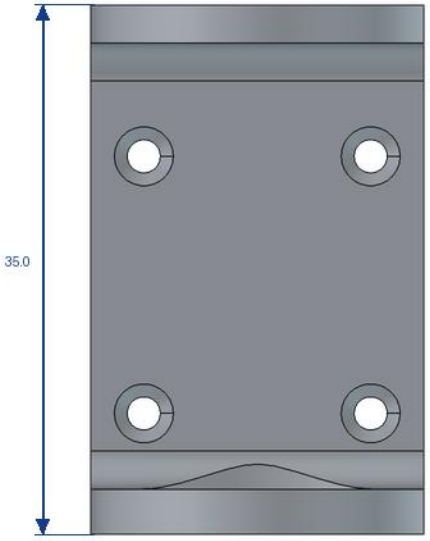
Olculer mm. Genel zarf olculeri verilmistir; delikler ve detaylar icin STEP/STL kullanin.



ON GORUNUS



SOLDAN GORUNUS



USTTEN GORUNUS

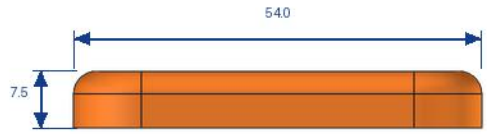


OpenDuck

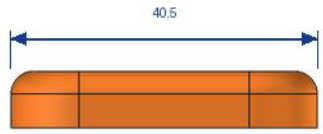
Microduck benzeri ordek robot

PARCA	ankle_bracket
ACIKLAMA	ayak bilegi U braketi
ADET	2 (sol + sag, ayna)
MALZEME	PETG (veya PLA+), renk: gri
BASKI	0.2 mm katman, 4 duvar, %40 dolgu
KUTLE	5.0 g (adet basi, tahmini)
ZARF	22.0 x 35.0 x 27.0 mm
YON	otomatik (genis temas, az destek)
DESTEK	gerekmez
OLCEK	2:1 (A4, 150 dpi)
IZDUSUM	ISO E - 1. aci, baski yonunde
DOSYA	left_ankle_bracket.stl, right_ankle_bracket.stl

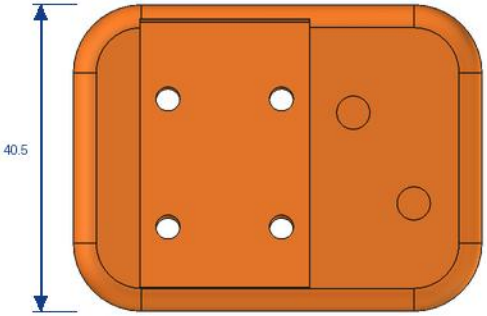
Olculer mm. Genel zarf olculeri verilmistir; delikler ve detaylar icin STEP/STL kullanin.



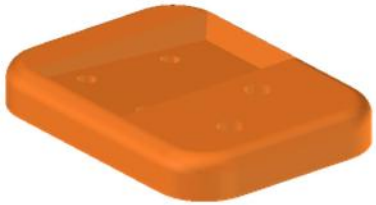
ON GORUNUS



SOLDAN GORUNUS



USTTEN GORUNUS

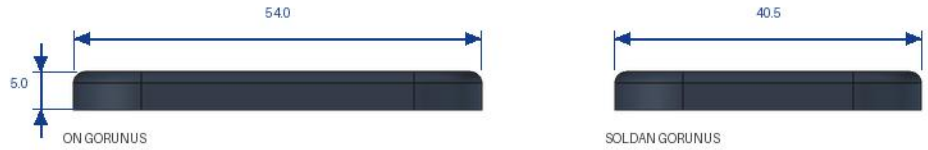


OpenDuck

Microduck benzeri ordek robot

PARCA	foot_body
ACIKLAMA	ayak govdesi
ADET	2 (sol + sag, ayna)
MALZEME	PLA, renk: turuncu
BASKI	0.2 mm katman, 3 duvar, %20 dolgu
KUTLE	13.8 g (adet basi, tahmini)
ZARF	54.0 x 40.5 x 7.5 mm
YON	otomatik (genis temas, az destek)
DESTEK	gerekmez
OLCEK	1:1 (A4, 150 dpi)
IZDUSUM	ISO E - 1. aci, baski yonunde
DOSYA	left_foot_body.stl, right_foot_body.stl

Olculer mm. Genel zarf olculeri verilmistir; delikler ve detaylar icin STEP/STL kullanin.



ON GORUNUS

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USTTEN GORUNUS

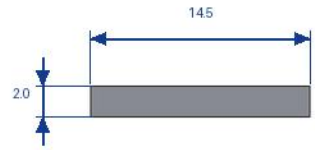


OpenDuck

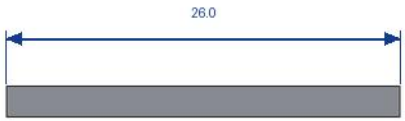
Microduck benzeri ordek robot

PARCA	sole
ACIKLAMA	taban (TPU)
ADET	2 (sol + sag, ayna)
MALZEME	TPU 95A, renk: koyu
BASKI	0.2 mm katman, 3 duvar, %30 dolgu, yavas
KUTLE	12.6 g (adet basi, tahmini)
ZARF	54.0 x 40.5 x 5.0 mm
YON	otomatik (genis temas, az destek)
DESTEK	gerekmez
OLCEK	1:1 (A4, 150 dpi)
IZDUSUM	ISO E - 1. aci, baski yonunde
DOSYA	left_sole.stl, right_sole.stl

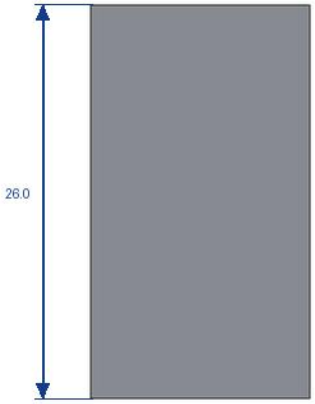
Olculer mm. Genel zarf olculeri verilmistir; delikler ve detaylar icin STEP/STL kullanin.



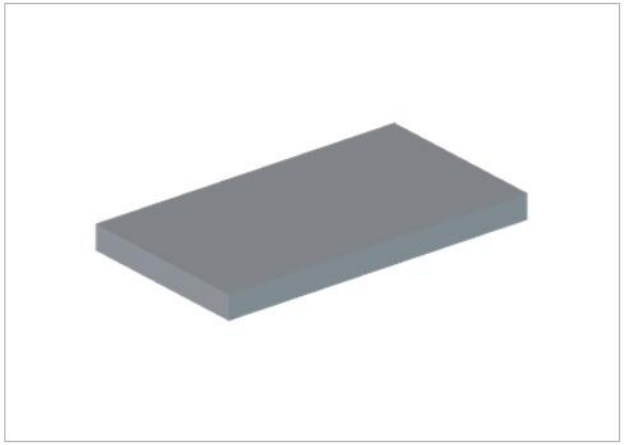
ON GORUNUS



SOLDAN GORUNUS



USTTEN GORUNUS

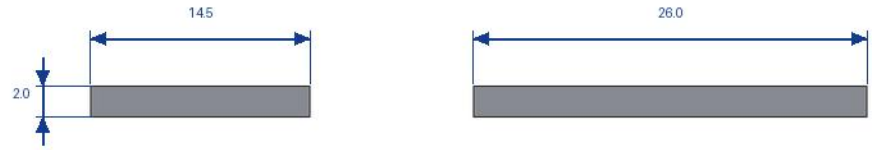


OpenDuck

Microduck benzeri ordek robot

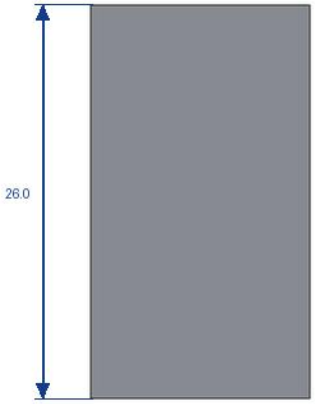
PARCA	neck_splice_1
ACIKLAMA	boyun ek plakalari (ayri parca 1/2)
ADET	1
MALZEME	PETG (veya PLA+), renk: gri
BASKI	0.2 mm katman, 4 duvar, %40 dolgu
KUTLE	0.8 g (adet basi, tahmini)
ZARF	14.5 x 26.0 x 2.0 mm
YON	duz plaka, tablada
DESTEK	gerekmez
OLCEK	2:1 (A4, 150 dpi)
IZDUSUM	ISO E - 1. aci, baski yonunde
DOSYA	neck_splice_1.stl

Olculer mm. Genel zarf olculeri verilmistir; delikler ve detaylar icin STEP/STL kullanin.

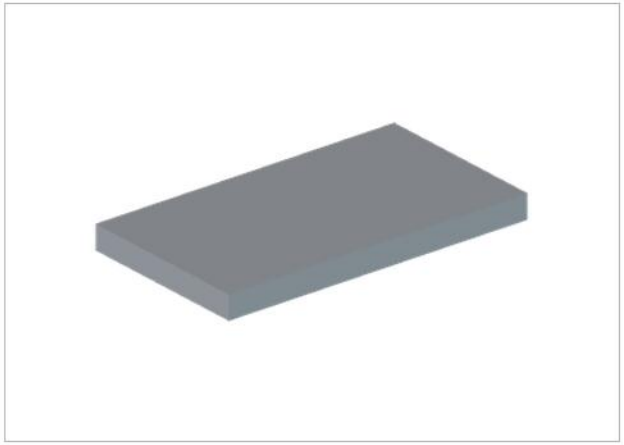


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USTTEN GORUNUS

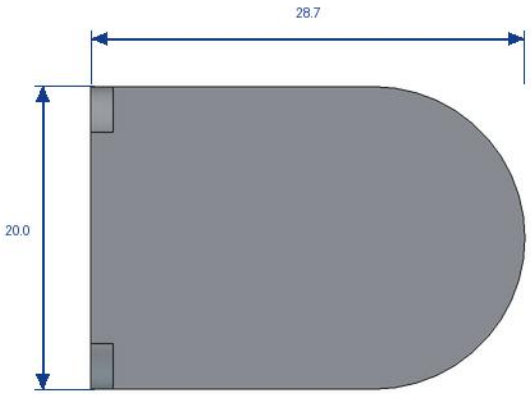


OpenDuck

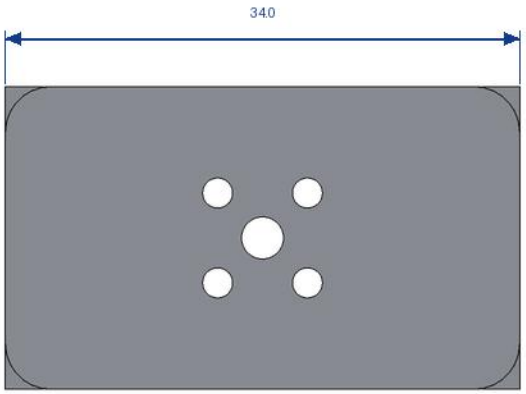
Microduck benzeri ordek robot

PARCA	neck_splice_2
ACIKLAMA	boyun ek plakalari (ayri parca 2/2)
ADET	1
MALZEME	PETG (veya PLA+), renk: gri
BASKI	0.2 mm katman, 4 duvar, %40 dolgu
KUTLE	0.8 g (adet basi, tahmini)
ZARF	14.5 x 26.0 x 2.0 mm
YON	duz plaka, tablada
DESTEK	gerekmez
OLCEK	2:1 (A4, 150 dpi)
IZDUSUM	ISO E - 1. aci, baski yonunde
DOSYA	neck_splice_2.stl

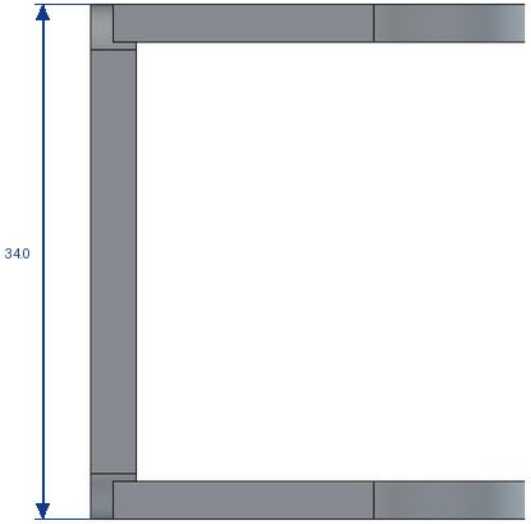
Olculer mm. Genel zarf olculeri verilmistir; delikler ve detaylar icin STEP/STL kullanin.



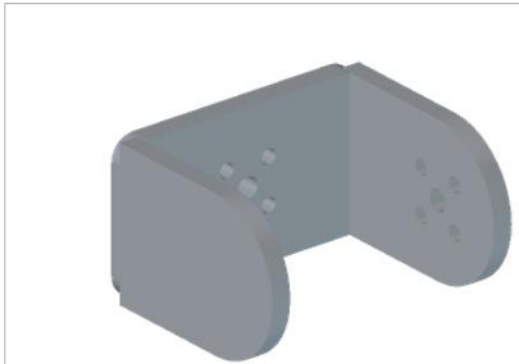
ON GORUNUS



SOLDAN GORUNUS



USTTEN GORUNUS



OpenDuck

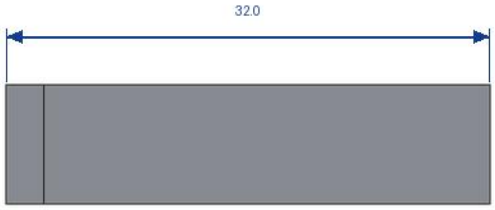
Microduck benzeri ordek robot

PARCA	head_tilt_bracket
ACIKLAMA	kafa egim braketi
ADET	1
MALZEME	PETG (veya PLA+), renk: gri
BASKI	0.2 mm katman, 4 duvar, %40 dolgu
KUTLE	4.7 g (adet basi, tahmini)
ZARF	28.7 x 34.0 x 20.0 mm
YON	U profili katman duzleminde (kafa egim torku katman arasina binmez)
DESTEK	gerekmez
OLCEK	2:1 (A4, 150 dpi)
IZDUSUM	ISO E - 1. aci, baski yonunde
DOSYA	head_tilt_bracket.stl

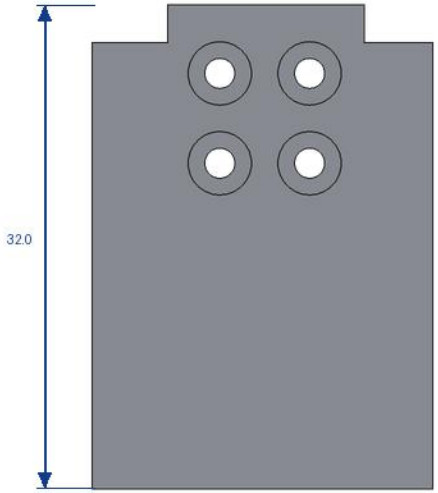
Olculer mm. Genel zarf olculeri verilmistir; delikler ve detaylar icin STEP/STL kullanin.



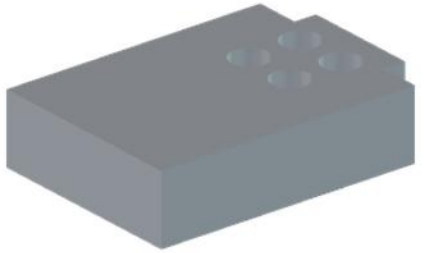
ON GORUNUS



SOLDAN GORUNUS



USTTEN GORUNUS

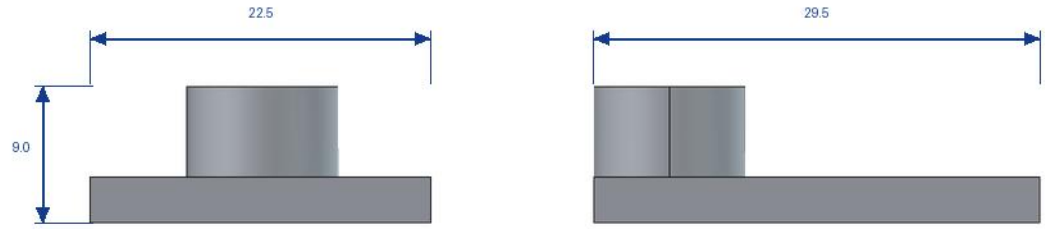


OpenDuck

Microduck benzeri ordek robot

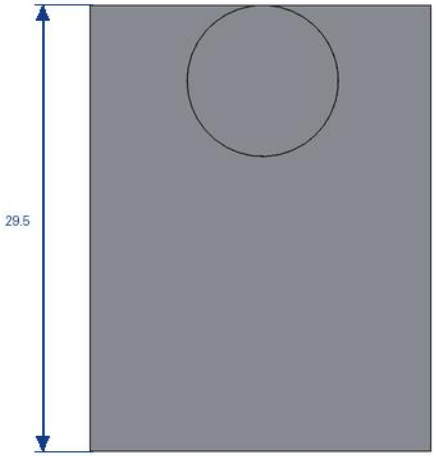
PARCA	head_turn_bracket_1
ACIKLAMA	kafa donus braketi (ayri parca 1/2)
ADET	1
MALZEME	PETG (veya PLA+), renk: gri
BASKI	0.2 mm katman, 4 duvar, %40 dolgu
KUTLE	5.8 g (adet basi, tahmini)
ZARF	22.5 x 32.0 x 7.9 mm
YON	-X yuzu tablada, 10 mm pim yukari bakar ve desteksiz yuvarlak basilir
DESTEK	gerekmez
OLCEK	2:1 (A4, 150 dpi)
IZDUSUM	ISO E - 1. aci, baski yonunde
DOSYA	head_turn_bracket_1.stl

Olculer mm. Genel zarf olculeri verilmistir; delikler ve detaylar icin STEP/STL kullanin.

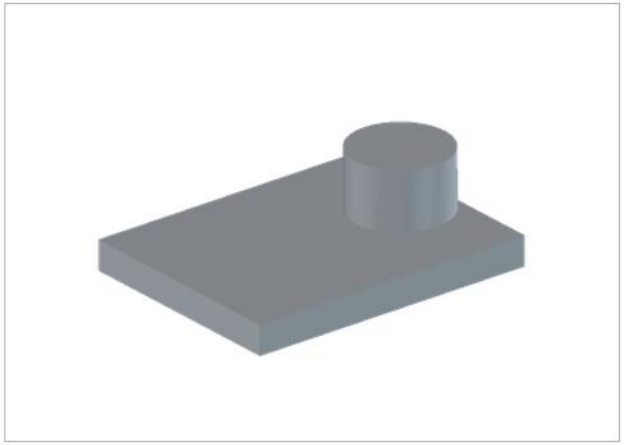


ON GORUNUS

SOLDAN GORUNUS



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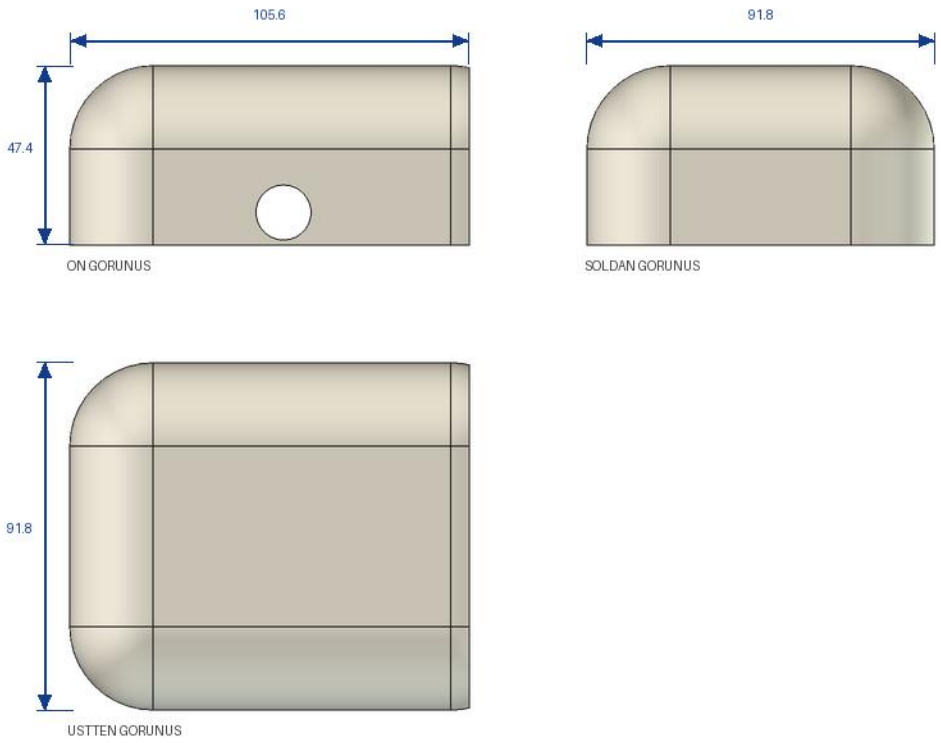


OpenDuck

Microduck benzeri ordek robot

PARCA	head_turn_bracket_2
ACIKLAMA	kafa donus braketi (ayri parca 2/2)
ADET	1
MALZEME	PETG (veya PLA+), renk: gri
BASKI	0.2 mm katman, 4 duvar, %40 dolgu
KUTLE	2.7 g (adet basi, tahmini)
ZARF	22.5 x 29.5 x 9.0 mm
YON	-X yuzu tablada, 10 mm pim yukari bakar ve desteksiz yuvarlak basilir
DESTEK	gerekmez
OLCEK	2:1 (A4, 150 dpi)
IZDUSUM	ISO E - 1. aci, baski yonunde
DOSYA	head_turn_bracket_2.stl

Olculer mm. Genel zarf olculeri verilmistir; delikler ve detaylar icin STEP/STL kullanin.



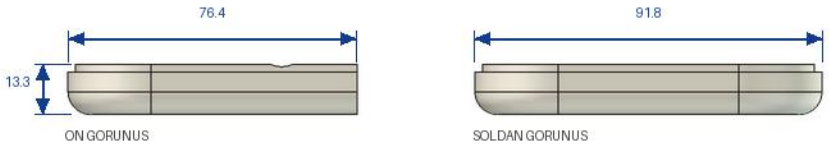


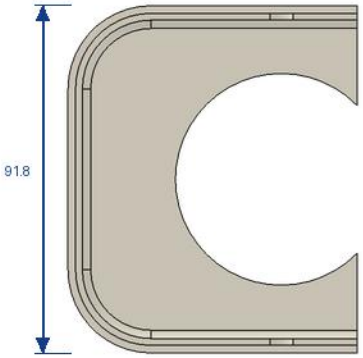
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Microduck benzeri ordek robot

PARCA	head_hood
ACIKLAMA	kafa ust kaput
ADET	1
MALZEME	PLA, renk: krem
BASKI	0.2 mm katman, 3 duvar, %15 dolgu
KUTLE	38.1 g (adet basi, tahmini)
ZARF	105.6 x 91.8 x 47.4 mm
YON	acik alt tablada: dis yuzeyde destek yok, 47 mm (sirtustu 109 mm); ic tavana tabladan agac (tree) destek
DESTEK	gerekli (tabladan)
OLCEK	1:2 (A4, 150 dpi)
IZDUSUM	ISO E - 1. aci, baski yonunde
DOSYA	head_hood.stl

Olculer mm. Genel zarf olculeri verilmistir; delikler ve detaylar icin STEP/STL kullanin.





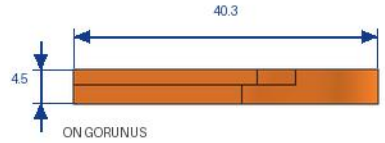


OpenDuck

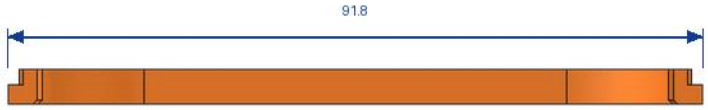
Microduck benzeri ordek robot

PARCA	head_bottom_shell
ACIKLAMA	kafa alt kabugu
ADET	1
MALZEME	PLA, renk: krem
BASKI	0.2 mm katman, 3 duvar, %15 dolgu
KUTLE	14.5 g (adet basi, tahmini)
ZARF	76.4 x 91.8 x 13.3 mm
YON	otomatik (genis temas, az destek)
DESTEK	gerekmez
OLCEK	1:2 (A4, 150 dpi)
IZDUSUM	ISO E - 1. aci, baski yonunde
DOSYA	head_bottom_shell.stl

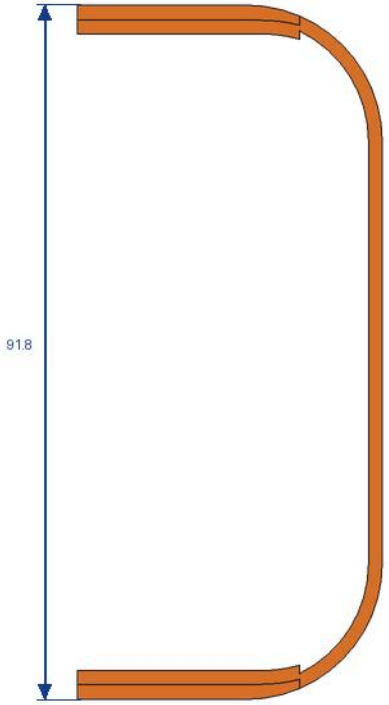
Olculer mm. Genel zarf olculeri verilmistir; delikler ve detaylar icin STEP/STL kullanin.



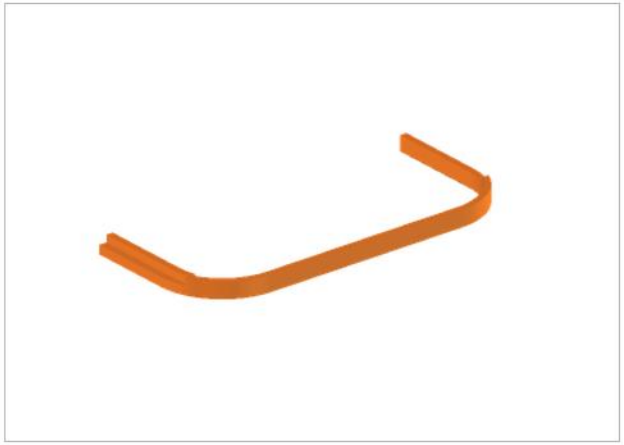
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Microduck benzeri ordek robot

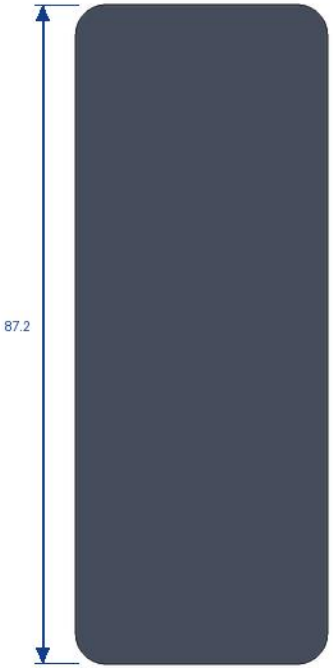
PARCA	upper_bill
ACIKLAMA	ust gaga
ADET	1
MALZEME	PLA, renk: turuncu
BASKI	0.2 mm katman, 3 duvar, %20 dolgu
KUTLE	1.7 g (adet basi, tahmini)
ZARF	40.3 x 91.8 x 4.5 mm
YON	otomatik (genis temas, az destek)
DESTEK	gerekmez
OLCEK	1:1 (A4, 150 dpi)
IZDUSUM	ISO E - 1. aci, baski yonunde
DOSYA	upper_bill.stl

Olculer mm. Genel zarf olculeri verilmistir; delikler ve detaylar icin STEP/STL kullanin.



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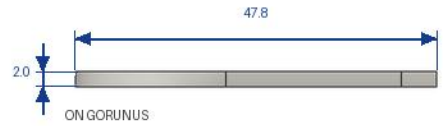


OpenDuck

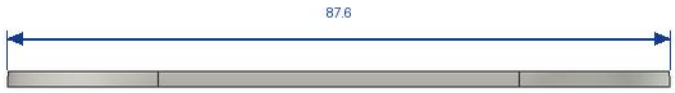
Microduck benzeri ordek robot

PARCA	upper_bill_pad
ACIKLAMA	ust gaga pedi (TPU)
ADET	1
MALZEME	TPU 85-95A, renk: koyu
BASKI	0.2 mm katman, %100 dolgu, yavas
KUTLE	3.8 g (adet basi, tahmini)
ZARF	33.3 x 87.2 x 1.1 mm
YON	otomatik (genis temas, az destek)
DESTEK	gerekmez
OLCEK	1:1 (A4, 150 dpi)
IZDUSUM	ISO E - 1. aci, baski yonunde
DOSYA	upper_bill_pad.stl

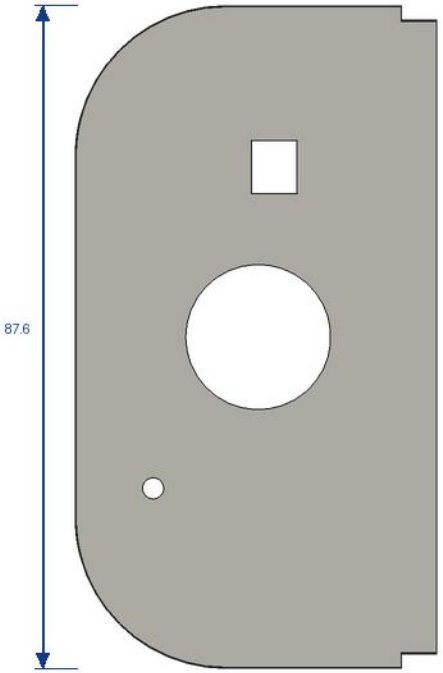
Olculer mm. Genel zarf olculeri verilmistir; delikler ve detaylar icin STEP/STL kullanin.



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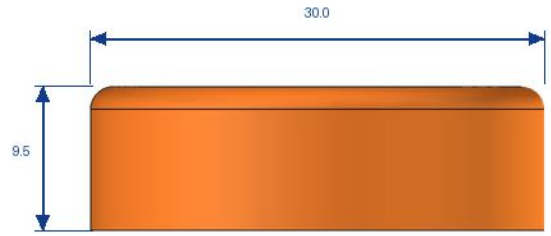


OpenDuck

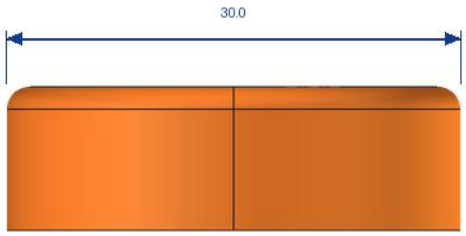
Microduck benzeri ordek robot

PARCA	face_plate
ACIKLAMA	yuz plakasi
ADET	1
MALZEME	PLA, renk: acik gri
BASKI	0.2 mm katman, 3 duvar, %20 dolgu
KUTLE	8.0 g (adet basi, tahmini)
ZARF	47.8 x 87.6 x 2.0 mm
YON	otomatik (genis temas, az destek)
DESTEK	gerekmez
OLCEK	1:1 (A4, 150 dpi)
IZDUSUM	ISO E - 1. aci, baski yonunde
DOSYA	face_plate.stl

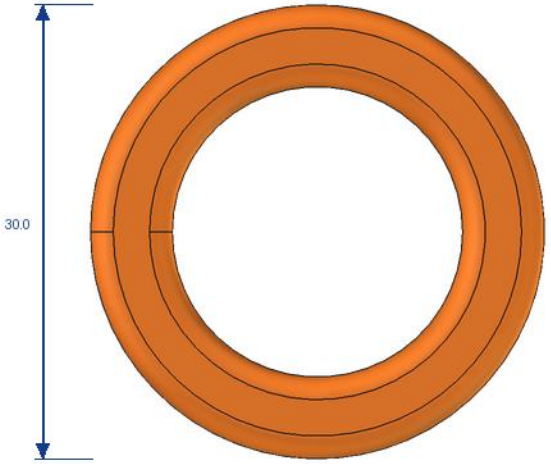
Olculer mm. Genel zarf olculeri verilmistir; delikler ve detaylar icin STEP/STL kullanin.



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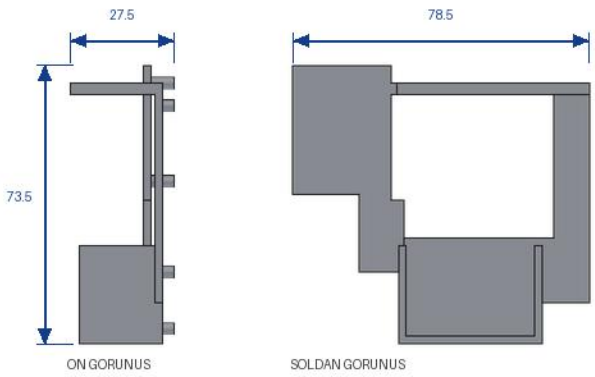


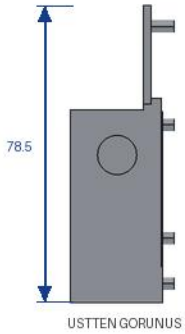
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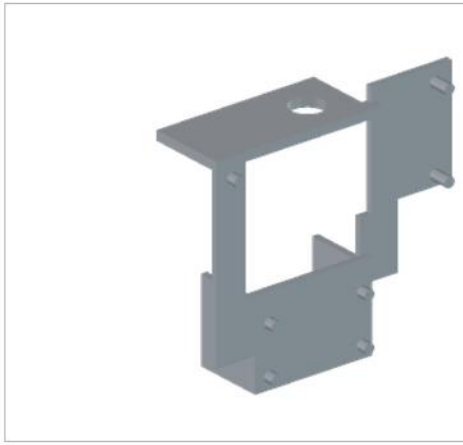
Microduck benzeri ordek robot

PARCA	eye_ring
ACIKLAMA	goz halkasi
ADET	1
MALZEME	PLA, renk: turuncu
BASKI	0.2 mm katman, 3 duvar, %20 dolgu
KUTLE	4.3 g (adet basi, tahmini)
ZARF	30.0 x 30.0 x 9.5 mm
YON	otomatik (genis temas, az destek)
DESTEK	gerekmez
OLCEK	2:1 (A4, 150 dpi)
IZDUSUM	ISO E - 1. aci, baski yonunde
DOSYA	eye_ring.stl

Olculer mm. Genel zarf olculeri verilmistir; delikler ve detaylar icin STEP/STL kullanin.





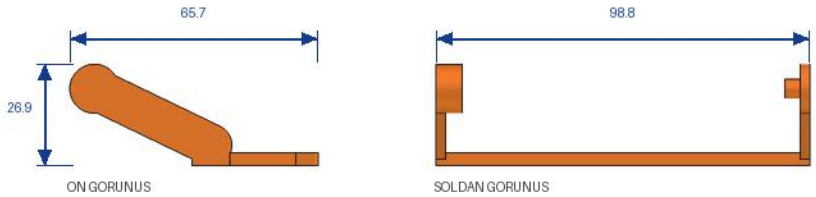


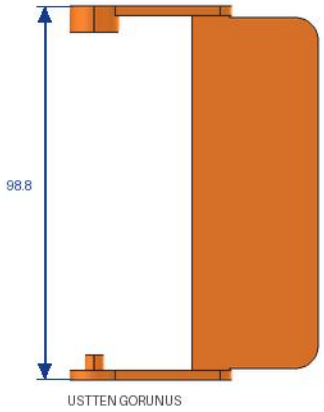
OpenDuck

Microduck benzeri ordek robot

PARCA	head_frame
ACIKLAMA	kafa ic cercevesi
ADET	1
MALZEME	PETG (veya PLA+), renk: gri
BASKI	0.2 mm katman, 4 duvar, %40 dolgu
KUTLE	13.9 g (adet basi, tahmini)
ZARF	27.5 x 78.5 x 73.5 mm
YON	otomatik (genis temas, az destek)
DESTEK	gerekli (tabladan)
OLCEK	1:2 (A4, 150 dpi)
IZDUSUM	ISO E - 1. aci, baski yonunde
DOSYA	head_frame.stl

Olculer mm. Genel zarf olculeri verilmistir; delikler ve detaylar icin STEP/STL kullanin.







OpenDuck

Microduck benzeri ordek robot

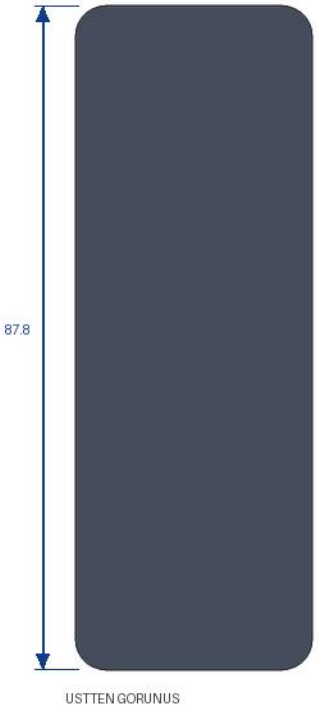
PARCA	lower_bill
ACIKLAMA	alt gaga (cene)
ADET	1
MALZEME	PLA, renk: turuncu
BASKI	0.2 mm katman, 3 duvar, %20 dolgu
KUTLE	14.9 g (adet basi, tahmini)
ZARF	65.7 x 98.8 x 26.9 mm
YON	otomatik (genis temas, az destek)
DESTEK	gerekli (tabladan)
OLCEK	1:2 (A4, 150 dpi)
IZDUSUM	ISO E - 1. aci, baski yonunde
DOSYA	lower_bill.stl

Olculer mm. Genel zarf olculeri verilmistir; delikler ve detaylar icin STEP/STL kullanin.

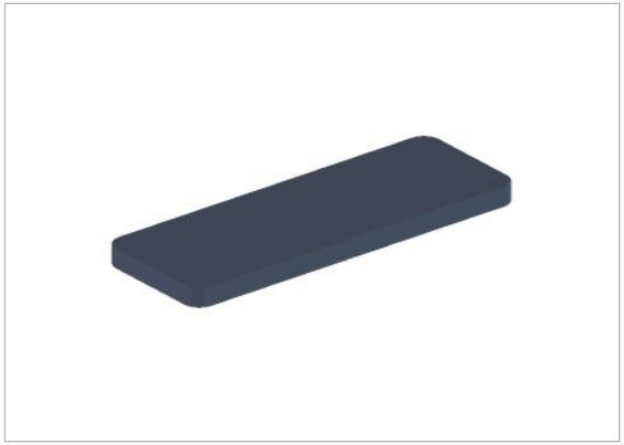


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Microduck benzeri ordek robot

PARCA	lower_bill_pad
ACIKLAMA	alt gaga pedi (TPU)
ADET	1
MALZEME	TPU 85-95A, renk: koyu
BASKI	0.2 mm katman, %100 dolgu, yavas
KUTLE	15.8 g (adet basi, tahmini)
ZARF	31.3 x 87.8 x 4.8 mm
YON	otomatik (genis temas, az destek)
DESTEK	gerekmez
OLCEK	1:1 (A4, 150 dpi)
IZDUSUM	ISO E - 1. aci, baski yonunde
DOSYA	lower_bill_pad.stl